Blessed G/Michael Catholic Secondary School

Semester 1 Biology Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which organelle is considered the "powerhouse of the cell" responsible for ATP synthesis?

A. Golgi apparatus **B.** Ribosome

C. Mitochondrion **D.** Lysosome

____ **2.** The movement of water molecules from a region of lower solute concentration to higher solute concentration across a semi-permeable membrane is:

A. Active transport **B.** Osmosis

C. Facilitated diffusion **D.** Endocytosis

____ **3.** During which phase of mitosis do sister chromatids separate and move toward opposite poles?

A. Prophase **B.** Metaphase **C.** Anaphase **D.** Telophase

____ **4.** In which stage of the cell cycle is DNA duplicated?

A. G1 phase **B.** S phase **C.** G2 phase **D.** M phase

____ **5.** Which biological branch specifically studies heredity and variation in living organisms?

A. Ecology **B.** Genetics **C.** Cytology **D.** Physiology

____ **6.** Which structure is present in plant cells but absent in typical animal cells?

A. Cell wall and chloroplast **B.** Ribosome and nucleus

C. Plasma membrane and cytoplasm **D.** Mitochondria and vacuole

____ **7.** Which of the following microscopes uses beams of electrons rather than light to achieve magnifications up to 1,000,000x?

A. Compound light microscope **B.** Transmission electron microscope (TEM)

C. Dissecting stereomicroscope **D.** Fluorescence microscope

____ **8.** What happens to a red blood cell placed into a hypertonic salt solution?

A. It absorbs water and lyses (bursts) **B.** It loses water and crenates (shrinks)

C. It remains unchanged in shape **D.** It divides by mitosis

____ **9.** Which component of the cytoskeleton forms the mitotic spindle fibers during nuclear cell division?

A. Microfilaments (actin) **B.** Intermediate filaments

C. Microtubules (tubulin) **D.** Collagen fibers

____ **10.** What is the end product of normal mitotic cell division?

A. Four genetically distinct haploid gametes
B. Two genetically identical diploid daughter cells
C. Two haploid zygotes
D. One multinucleated clone